

Piyush Kumar Arun

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EDUCATION

National Institute of Technology Calicut

M.Tech in Computer Science & Engineering

- Relevant Courses: Information Security, AI Agents, Distributed Systems, Linear Algebra, Big Data

University of Delhi

B.Tech in IT and Mathematical Innovation (CGPA: 8.85/10)

- Relevant Courses: Data Structures and Algorithm, DBMS, Object-Oriented Programming, Operating System, SDLC

August 2024 – May 2026

Calicut, India

August 2019 – July 2023

Delhi, India

WORK EXPERIENCE

Tazapay

AI Agent Intern (LLMs, Prompt Engineering, n8n, automation, AI Agent)

Bengaluru, India

- AI automation & n8n orchestration:** Architected and deployed production-grade workflow automation using n8n, integrating webhook APIs, OAuth2 authentication, idempotency controls, data validation, third-party bank APIs, and Slack monitoring to enable secure, scalable, and fully automated payout processing.
- Cross-border transaction time prediction:** Built a production-ready payout time prediction model for cross-border transactions to India using SQL-based data analysis, statistical metrics (mean, median, quantiles), feature engineering (currency, amount bands, bank codes, compliance flags), and historical transaction datasets to accurately estimate and display expected settlement times to merchants at payout initiation.

Instantpay India Ltd

Software Engineer (FinTech, Angular, TypeScript, CSS, GitHub)

New Delhi, India

- Product Development & User Impact:** Developed and deployed [financial calculators](#) (EMI, TDS, GST, CAGR, Income Tax, etc.), improving user financial planning experience by 30% and contributing to overall platform engagement.
- High-Scale Systems & Reliability:** Developed secure internal Products (Expense Card, Gift Card, SFTP) capable of handling 1M+ daily transactions and \$1B+ monthly volume, ensuring high scalability & data security.
- Frontend Migration & Developer Efficiency:** Spearheaded frontend migration to Angular CLI v15+, resolving 30+ UI/UX issues, and managing GitHub repos through structured branching strategies and thorough code reviews.

PROJECTS

COM: Framework for Hallucination Reduction and Reasoning Enhancement of LLMs in Zero-Shot Code Generation | [Link](#)

- Developed a **POC orchestration workflow** using n8n to validate architectural feasibility, agent communication, and scalability, enabling rapid experimentation with agent roles, **prompt engineering**, and execution pipelines.
- Architected the **COM multi-agent framework** (Creator, Critic, Verifier, Refiner) integrating **domain-specific open-source LLMs**, achieving **~35% hallucination reduction** and improved **Pass@1 accuracy** in code generation.
- Evaluated the agentic self-correction pipeline on **HumanEval** and **GenEval**, improving logical consistency and reducing reasoning errors through structured critique and verification. (LLMs, GenAI, Multi-Agent, NLP)

AI Hybrid Personalized Travel Assistant | [Link](#)

- Designed and implemented a **hybrid retrieval system** using **Pinecone (vector similarity search)** and **Neo4j (graph-based queries)** to generate context-aware travel recommendations by combining semantic relevance with structured relationship reasoning.
- An **end-to-end AI system** integrating **Hugging Face LLMs & Gemini 2.5 pro**, async query execution, caching, and **LangChain-based conversation memory**, enabling multi-turn, stateful user interactions with improved performance and scalability. (Python, RAG, Vector & Graph Database, LangChain, Huggingface)

SKILLS

Languages: Python, C++, HTML, CSS | **Frameworks/Libraries:** LangChain, LangGraph, Angular | **AI/ML:** LLMs, Transformers, GenAI, n8n, RAG, Vector/Graph Databases, Agentic AI | **Tools:** Github, Git | **Practices:** Agile Development, Research, AWS

OTHER ACHIEVEMENTS

- Masters' research work has been accepted at **IndoML'25 & RECCAP'26**, will be published in **IEEE Proceedings**.
- Alumni Representative (Coordinator)**, CSE Dept., NIT Calicut – Demonstrated leadership & engagement initiatives.
- Secured **All India Rank - 67** in Delhi University Entrance Test (NTA DUET).